SWAIRA RIAZ

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SUMMARY

AI Engineer with hands-on experience building and deploying production AI systems, including RAG-powered chatbots, deep learning models, and real-time ML pipelines. Currently developing and maintaining an AI chatbot using RAG, Azure OpenAI, and workflow automation. Holds a BSc in Computer Systems Engineering, with a final year project on real-time network intrusion detection using LSTM neural networks. Skilled in Python, TensorFlow, and end-to-end deployment across cloud platforms.

EDUCATION

Bachelor of Science in Computer Systems Engineering
IslamiaUniversity of Bahawalpur | 2022-2026
Relevant Coursework: Machine Learning, Data Structures, Algorithms, Database Systems, Software Engineering

TECHNICAL SKILLS

- **Programming & Development:** Proficient in Python with expertise in algorithm development, data structures, and software engineering principles
- **AI/ML & Deep Learning:** TensorFlow, Keras, Scikit-learn, LSTM, CNN, ResNet, Transfer Learning, XGBoost
- **Generative AI & Chatbots:** RAG Architecture, Azure OpenAI, Hugging Face Transformers, ChromaDB, Prompt Engineering
- **Automation & Integrations:** Zapier, Google Apps Script, Google Sheets Automation
- **Deployment & Cloud:** Docker, Kubernetes, FastAPI, Flask, Railway, Vercel, Firebase
- **Data Science & Visualization:** Pandas, NumPy, Matplotlib, Scapy, OpenCV

EXPERIENCE & COLLABORATION

- **AI Engineer — France-based Private Company | 2026 – Present**
  Developed, deployed, and maintained a production-grade AI chatbot using RAG architecture and Azure OpenAI. Automated business workflows using Zapier, Google Sheets, and Google Apps Script for data processing and system integrations. Deployed and maintained chatbot infrastructure on Railway, Vercel, and Firebase, ensuring scalable and reliable production performance.
- **IEEEVolunteer- GraphicDesigner | 2022-2024**
  Designed 15+ visual assets for IEEE technical events and academic conferences
  Collaborated across 5+ cross-functional team projects, demonstrating leadership and communication

PROJECTS

Cyber Twin: AI-Powered Network Intrusion Detection (Final Year Project)
Technologies: Python, TensorFlow, LSTM, React.js, FastAPI, Scapy

Built an LSTM-based intrusion detection system achieving 91.7% accuracy (97% DoS/DDoS recall) on the CICIDS-2017 dataset (385,000+ samples)
Developed a live packet-capture and flow-analysis pipeline with Scapy, extracting 25 real-time network features per flow. Built a real-time React.js dashboard streaming live alerts and network topology via WebSocket

AI-Powered Skincare Recommendation System
Technologies: Python, FastAPI, ChromaDB, ResNet, Docker, Kubernetes, RAG

Architected a ResNet-based image classifier achieving 92% accuracy on 10,000+ skin images
Implemented a RAG pipeline with ChromaDB embeddings processing 500+ skincare ingredients
Deployed a containerized FastAPI application with Docker, Kubernetes, and MLflow monitoring

Medicine Recommendation System
Technologies: Python, Flask, Scikit-learn, SVM, Pandas

Engineered an SVM classifier achieving 89% diagnostic accuracy across 15+ medical conditions
Built a recommendation engine providing medications, diet plans, and workout suggestions